Literature Review – Comparison Table

Robust Reconstruction of Tokamak Plasma State from Sparse Diagnostics Using Continuous-Time Neural Networks

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Papers are grouped into four threads: **(A)** reconstruction on real tokamak data – our dataset, task, and baselines; **(B)** robustness to missing/corrupted sensors – documented, but only for a forecasting task; **(C)** continuous-time models in tokamak research – applied to dynamics/forecasting, with spatial reconstruction appearing underexplored; **(D)** physics-informed / alternative reconstruction approaches and broader context.

Table 1: Literature review comparison table (15 papers, 4 threads + method background).

Paper	Yr.	Problem	Dataset	Method	Key Result	Limitation	Relevance to Our Work
Thread A – Reconstruction on real tokamak data (our dataset, task, and baselines)							
Rousseau et al., TokaMark, arXiv:2602.10132v3 (Table 3)	2026	Standardized benchmark: reconstruct flux map / plasma boundary from sparse magnetics; 14 tasks total	11,573 real MAST tokamak shots (FAIR-MAST), public S3	CNN and CNN+LSTM baselines per task	Task 1-3 (flux map): CNN+LSTM 0.159 NRMSE (random) → 0.257 (temporal); CNN 0.148 → 0.502 (v3, Table 3)	No robustness-to-corruption evaluation; numbers have shifted between paper revisions	**Our dataset, task definition, and baseline to reproduce and beat**
Boschi et al., TokaMind, arXiv:2602.15084	2026	Multi-modal transformer foundation model for all TokaMark tasks	Same (FAIR-MAST)	Transformer with masked/chunked tokenization, optional token/chunk-dropout reg. (§5.1)	Outperforms strongest baseline on all but Task 4-5 (Task 1-3 FT-Base 0.122); optional token/chunk dropout (§5.1)	Reports only clean-data accuracy; no systematic test-time corruption benchmark despite training with dropout	Strongest published baseline for our task; the robustness gap remains even for this model
Neural network reconstruction of the DIII-D plasma boundary using reduced diagnostics, J. Plasma Phys.	2026	Plasma boundary reconstruction from a reduced diagnostic set	Real DIII-D shots	0.03–0.04 m boundary error using coil currents alone vs. coil+current+voltage	Compares two *fixed* diagnostic sets chosen at design time, not corruption of a trained model at test time	Confirms sparse-sensor reconstruction works in-distribution on a second real tokamak	
Jalalvand et al., Multimodal Super-Resolution, Nat. Commun.	2025	Reconstruct a missing/degraded diagnostic using other available diagnostics	Real DIII-D data	Cross-diagnostic reconstruction recovers information from failed/absent sensors	Imputation-style cross-modal fix, not an architectural robustness test	Confirms handling degraded diagnostics is an active, current research direction	
Thread B – Robustness to missing/corrupted sensors (documented, but not for reconstruction)							
Gupta, Benchmarking Sensor Robustness in Plasma Diagnostic Models, arXiv:2607.11915	2026	Systematic robustness benchmark: train clean, corrupt test set under 6 scenarios	TokaMark (Task 4-4, *forecasting* – future plasma current)	LSTM: +212% NRMSE, disruption-alarm TPR collapses to 0% under event-proximate sensor failure	**Tested only on a forecasting task, never on reconstruction (Task 1-2/1-3)**; solo-authored preprint, not yet peer-reviewed	**Our methodology template** – we extend this exact procedure to reconstruction	
Yang et al., FusionMAE, arXiv:2509.12945	2025	Unify diagnostics into one embedding; reconstruct masked/missing channels	HL-3 tokamak, 2,445 discharges, 88 channels	96.7% reliability reconstructing masked channels	1D signal imputation via masking, not 2D spatial-field reconstruction quality	A data-level (masking) fix for missing data; contrasts with our architectural (continuous-time) approach	
Thread C – Continuous-time models in tokamak research (dynamics, not reconstruction)							
Application of Neural ODEs for Tokamak Plasma Dynamics Analysis, arXiv:2403.01635	2024	Model plasma transport dynamics (core/edge/SOL energy exchange)	Real DIII-D data	Neural ODE fits transport/diffusivity dynamics	0D physics-parameter fitting, not spatial reconstruction from sensors	Shows Neural ODEs are usable on real tokamak data, but for a different problem class	

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Hybridizing Physics and Neural ODEs for Plasma Inductance Dynamics, NeurIPS workshop	2023	Forecast plasma current and internal inductance (scalar time series)	Alcator C-Mod	Physics-informed Neural ODE forecasts scalar dynamics	Scalar forecasting, not spatial field reconstruction	Further confirms Neural ODE use is limited to dynamics/forecasting in this domain	
Arnold, Spangher & Rea, Continuous CNN for Disruption Prediction, arXiv:2312.01286	2023	Disruption prediction (binary classification) via continuous convolution kernels	Alcator C-Mod	Continuous CNN AUC 0.974 vs. discrete CNN AUC 0.799	Different continuous-time mechanism (kernels, not recurrent state); authors explicitly state robustness to missing data **not yet tested** (their own stated limitation)	Strongest evidence continuous-time models outperform discrete ones on real tokamak time series – but robustness-untested, and not applied to reconstruction	
Thread D – Physics-informed / alternative reconstruction approaches and context							
Fusion Equilibrium Challenge, arXiv:2609.01750	2026	Flux map reconstruction from non-magnetic diagnostics; cross-machine transfer	9,113 DIII-D + 2,416 MAST shots (open, Hugging Face)	PCA+Ridge / GBM / Conv-decoder baselines; SSIM 0.83 in-machine	Collapses to SSIM ∼0.10 cross-machine; treats each timestep independently, no temporal modeling at all	Alternative real dataset; cross-machine generalization is a different (also open) robustness question from ours	
Rutigliano et al., PINN Equilibrium + Kinetic Profile Reconstruction, Nucl. Fusion 66:046006	2026	Joint equilibrium + kinetic profile inverse problem	JET + TokaLab synthetic	Physics-informed NN, R^2 >99% synthetic	Heavy physics modeling, no dynamics, no robustness testing	Physics-informed alternative to our purely data-driven approach	
Joung et al., GS-DeepNet, Sci. Rep.	2023	Deep-learning surrogate for the Grad-Shafranov equilibrium equation	Simulated + real equilibria	Fast, accurate equilibrium surrogate	Physics-constrained surrogate, not a sparse-sensor learned mapping	Contrast case: physics-informed surrogate vs. our purely learned, sensor-driven approach	
Degrave et al., Magnetic Control of Tokamak Plasmas via Deep RL, Nature	2022	Real-time magnetic control of tokamak plasma shape	TCV (simulation-trained, deployed real-time)	Landmark demonstration of DL-based real-time tokamak control	Not a reconstruction task; assumes accurate state estimation as an input	Motivates why fast, accurate state reconstruction (our task) matters for control	

Method background (cited in text and in the main Mid-Review references, not tabulated above as they are not tokamak reconstruction papers): Neural ODEs (Chen et al., 2018); Latent ODEs (Rubanova et al., 2019); Liquid Time-Constant Networks (Hasani et al., 2021); Closed-form Continuous-time Networks (CfC; Hasani et al., *Nat. Mach. Intell.*, 2022) – these establish the continuous-time architectures (CfC/LTC) used as our proposed encoder.